CSE232 Labwork 1

Duration: 50 minutes

General Rules

- Write each task as a separate program ending with `.END`.
- Use `#` for immediate values and `$` for memory addresses.
- Put the test inputs into memory at the start of each task (`LDAA #value` / `STAA $addr`).
- Put all of your code into a text file and upload the text file as **name_surname_id.txt** to yulearn.

Task 0: Warm-up (15 points)

- Load decimal value 25 into Accumulator A and hexadecimal value FD into Accumulator B.
- Store Accumulator A at `$40` and Accumulator B at `$41`.
- Add the two values and store the result at `$45`.

Test	**Inputs**	**Expected (dec / hex)**
1	X = 25, Y = FD	22 / $16

Task 1: Arithmetic Translation (40 points)

Implement the expression below using memory inputs:

Z = (X - 1) * 6 − 2 * Y

Requirements

- Read X from `$50` and Y from `$51`.
- Store the final result Z at `$53`.
- You may use `$52` for intermediate results.
- Use shifts for all multiplications.

Test	Inputs	Expected (dec / hex)
1	X = 5, Y = 3	18 / $12
2	X = 2, Y = 10	-14 / $F2

Task 2: Absolute Difference (45 points)

Compute the difference of two numbers and make sure the stored result is positive:

Z = X − Y

Requirements

- Read X from `$60` and Y from `$61`.
- If Z is positive (or zero), store it directly at `$62`.
- If Z is negative, convert it to its positive value and store that at `$62`.
- `NEG` / `NEGA` / `NEGB` are not allowed.
- Hint: Use 2's complement to convert from negative to positive.

Test	Inputs	Expected (dec/hex)
1	X = 20, Y = 7	13 / $0D
2	X = 5, Y = 12	7 / $07